

LETTER TO THE EDITOR

Performance and accuracy of 10 AI-based algorithms in diabetic retinopathy screening in a real-world clinical setting

Editor,

Diabetic retinopathy (DR) is a leading cause of preventable blindness emerging as a global health challenge (Teo et al., 2021). As DR prevalence continues to rise, improvements in screening methods have become crucial for timely detection and intervention (Sun et al., 2022). The development of portable handheld fundus cameras may help to widen access to screening (Kubin et al., 2021). In addition, integration of AI algorithms in screening would significantly enhance its efficiency and help with the resource shortage with streamlined workflows (Xie et al., 2020).

We studied the performance of 25 commercially available AI algorithms to assess their potential impact on the screening workflow. A total of 624 images (macula- and papilla-centred 50°-images from each eye under mydriasis) were captured with a handheld camera, Optomed Aurora, from 156 consecutive patients with diabetes attending DR screening or follow-up in the Oulu University Hospital's outpatient eye clinic. Images were analysed independently and in high agreement by two ophthalmologists blinded to AI outcomes, and the ophthalmologist's grading for DR was used as a reference. DR was graded according to the Finnish DR screening programme: no DR, mild non-proliferative diabetic retinopathy (NPDR), moderate NPDR, severe NPDR and proliferative diabetic retinopathy (PDR). Here, we focus on the 10 best-performing algorithms when assessed by the agreement with ophthalmologist's grading, sensitivity and specificity. The selected algorithms were ranked by Youden's index (= Sensitivity + Specificity – 1). The

study followed the tenets of the Declaration of Helsinki, and it was approved by Oulu University Hospital Ethical Research Committee.

Non-referable DR (no DR or mild NPDR) was detected in 102 (65%) and referable DR (moderate or severe NPDR, and PDR) in 54 participants (35%). The highest agreements between human grading and AI were above 90%. The sensitivities and specificities of the algorithms were generally good compared to the screening recommendations, and the Youden's indexes ranged from 0.82 to 0.65 (Table 1). The number of false positives and negatives varied from 3 to 14 by individual algorithms. Most of the false positives were caused by retinal abnormalities other than DR, that is age-related macular degeneration, branch or central retinal vein occlusion, which also have clinical significance. Among 1560 analysis results from 10 different AI algorithms, 6 (0.4%) clinically severe false-negative cases were identified. These six severe cases were distributed across five algorithms: one AI missed two severe cases, while four others each missed one.

In general, in terms of high sensitivity and specificity, these selected algorithms performed very well in detecting DR in fundus images captured by a handheld fundus camera without any optimization or localization. These results indicate that numerous high-quality algorithm alternatives are currently in the market if one considers implementing AI in the DR screening workflow (Kubin et al., 2024). However, the validation/localization of the algorithm is necessary before taking any AI solution into clinical use, and this would most probably further

TABLE 1 The performance of the algorithms ranked by Youden's index.

AI	Agreement [%]	Sensitivity [%]	Specificity [%]	PPV [%]	NPV [%]	Youden's index
1	92.3	85.0	96.9	94.4	91.2	0.82
2	91.0	85.0	94.8	91.1	91.0	0.80
3	85.9	91.7	82.3	79.7	94.0	0.74
4a	87.8	76.7	94.8	90.2	86.7	0.71
4b	85.9	83.1	87.6	80.3	89.5	0.71
6a	85.3	81.7	87.5	80.3	88.4	0.69
6b	86.5	76.7	92.7	86.8	86.4	0.69
8	84.0	83.3	84.4	78.1	89.0	0.68
9a	82.1	83.3	81.3	84.3	88.6	0.65
9b	84.0	76.7	88.5	80.7	85.9	0.65
Average	86.5	82.3	89.1	84.6	89.1	

Abbreviations: AI, artificial intelligence; NPV, negative predictive value; PPV, positive predictive value.

improve their performance. The detection of ocular pathology other than DR by the AI algorithm might protect the patient from diagnostic failure of other sight-threatening disorders. Human oversight is necessary to accurately diagnose and consider the requirement for intervention or further examinations for DR, but also for ocular diseases beyond DR in screening data.

The implementation of effective AI may enhance the efficacy and accuracy of DR screening, potentially reducing unnecessary referrals to ophthalmic clinics by minimizing false-positive cases. In conclusion, the validated use of AI together with a high-quality portable fundus camera holds vast potential for streamlining the screening process, reducing the burden on healthcare professionals, and improving access to screening and patient outcomes by enhanced diagnostic accuracy.

FUNDING INFORMATION

The Finnish Eye Foundation, Grant/Award Number: 00082022.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interests in the present study. Petri Huhtinen and Laura Piila are employers of Optomed.

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Received: 16 February 2025 | Accepted: 17 June 2025

DOI: 10.1111/aos.17538

Comparative treatment burden of intravitreal injection versus visit frequency in patients with exudative age-related macular degeneration

Dear Editor,

The intravitreal injection (IVI) of anti-VEGF/angiopoietin-2 has revolutionized the therapy of neovascular age-related macular degeneration (nAMD), yet the associated treatment burden remains a pivotal concern in its long-term management. While previous studies have addressed treatment adherence and logistical challenges of IVI therapy (Boyle et al., 2017; Holekamp et al., 2024; Holz et al., 2020), data on the subjective burden related to the procedure and to follow-up visits are missing. Our

cross-sectional study aimed to close this gap by systematically comparing these two dimensions of treatment burden in a real-world setting.

We surveyed 305 patients with nAMD undergoing IVI therapy at a high-volume tertiary centre in Vienna, Austria. Inclusion required a minimum of five prior IVIs to ensure familiarity with treatment procedures. Between May 2023 and March 2024, patients completed a modified version of the LAF-IVT questionnaire (Holz et al., 2020), assessing treatment burden related to